A Two-Configuration Cavendish Test That Separates Low-Acceleration Gravity Models

A proposal for G metrology groups

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Abstract

Low-acceleration modifications of gravity make sharply different predictions for one inexpensive laboratory measurement. That measurement is the fractional change in the measured G when the gravitational acceleration that the source masses produce at the test masses is changed by a factor of four, with everything else held fixed.

- **Source-field-boost model.** One published model applies a MOND-type boost to the source field alone (Klein's interpolation, $\beta \simeq 1.3$, as used by Scarpa, Falomo & Treves 2026). It predicts a shift of about 340 ppm between configurations at half and twice the source acceleration of the Quinn et al. apparatus.
- **Dark-energy-tied model.** A framework in which the low-acceleration scale is tied to dark energy and screened at laboratory scales predicts no measurable shift: $|\Delta G/G| < 10^{-6}$, with a model value of order 10^{-20}.
- **Standard physics.** Newtonian gravity and general relativity predict zero.

Distinguishing the first prediction from the others needs a precision on the *difference* of about 30–50 ppm. Leading torsion-balance groups reach this today. We describe how to run the test on an existing apparatus: the two configurations, the systematics budget, blinding, and null checks. We also give the predictions in a form that can be checked directly.

1 The question and the competing predictions

- **The setup.** In a Cavendish-type experiment, the source masses produce an acceleration a_s at the test masses. Typically $a_s \sim 10^{-8}$–$10^{-7}\,\mathrm{m\,s^{-2}}$, far below Earth's $g \approx 9.8\,\mathrm{m\,s^{-2}}$ but still well above the galactic scale $a_0 \sim 10^{-10}\,\mathrm{m\,s^{-2}}$.
- **The observable.** Let $G(a_s)$ be the value an experiment infers.
- **The proposed measurement.** Measure

$$\delta \equiv \frac{G(a_1) - G(a_2)}{\bar{G}}, \qquad a_2/a_1 \approx 4,$$

on one instrument, with only the source configuration changed.

model	predicted δ (half → twice the nominal a_s)	reason
Newton / general relativity	0	G is a constant
Source-field boost (Klein, $\beta = 1.3$) [1]	$\approx 3.4 \times 10^{-4}$	boost $F(a_s/a_0)$ applie
Simple-ν boost on the source field alone (illustrative)	$\sim 3 \times 10^{-3}$	as above, steeper func
Dark-energy-tied, unscreened, Earth's field included [2]	$\sim 4 \times 10^{-20}$	Earth's g sets the reg
Dark-energy-tied, committed (stiffened) form [2,3]	0 (offset $\sim 10^{-36}$)	screened below parsec

Reproducing the source-boost numbers. Klein's function is $F(x) = [1 + x^{-\beta}]^{1/(2\beta)}$, with $x = a_s/a_0$, $a_0 = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $\beta = 1.3$. It reproduces the published predictions $G = 6.6769$ and 6.6747×10^{-11} (in SI units) for half and twice the Quinn source acceleration. That corresponds to $a_s \approx 4.7 \times 10^{-8}\,\mathrm{m\,s^{-2}}$ nominal, which is consistent with four 11 kg source masses at about 0.21 m. The difference is 338 ppm. (Lane CFG772 [2] reproduces both numbers to 8×10^{-6}.)

Why the dark-energy-tied framework predicts nothing. Three independent reasons, all in [2]:

- Earth's $g \gg a_0$ puts the whole laboratory deep in the Newtonian regime.
- The source masses are supported, not free-falling, so they do not define their own low-acceleration region.
- The committed form of the theory screens its extra field on scales below about 2 pc. This is required by the 2026 Cassini bound [4].

A source-dependent shift $|\delta| \geq 10^{-6}$ would falsify that framework.

2 How to run it on an existing apparatus

Any of the three standard readouts works: time-of-swing, angular acceleration feedback, or electrostatic servo. The proposal adds no new hardware beyond a second source configuration.

Choosing the two configurations

- **Option A, change the source mass (preferred).** Replace the source masses with a geometrically identical set of four times the density, or four times the mass at the same centre of mass. Examples: tungsten-alloy versus copper inserts, or hollow versus solid shells. Positions are unchanged, so the dominant geometric terms in the G extraction are unchanged. What is new is the density metrology (mass, homogeneity, dimensions) of the second set.
- **Option B, change the source distance.** Move the same masses so that a_s at the test masses changes by 4. This needs metrology of the new positions and a recomputed geometry model. Cross-check the multipole coefficients against a dedicated calibration run.
- **Option C, both.** Run A and B together and require the same G from all three to four configurations. Combining them separates any source-acceleration dependence from geometry errors.

Measurement plan

1. **Interleaving.** Alternate the two configurations in blocks (e.g. ABBA over days to weeks). This cancels slow drifts in fibre properties, temperature and tilt.
2. **Blinding.** An external party adds a hidden offset to the inferred G of each configuration. The offsets are removed only after the systematics budget is frozen.
3. **Null checks.** Rotate the whole source assembly and reverse its orientation. Use a "dummy" configuration with the same a_s but different masses: it must give $\delta = 0$. Vary the vacuum pressure and the temperature setpoint.
4. **Statistic.** Report δ with a full covariance, including correlated systematics between configurations.

Precision budget

- **Target.** $\sigma(\delta) \leq 50\,\text{ppm}$ separates the source-boost prediction from zero at $\geq 6\sigma$. $\sigma(\delta) \leq 30\,\text{ppm}$ gives $\geq 10\sigma$.
- **Why this is achievable.** The difference measurement inherits only the systematics that *change* between configurations: mass and density metrology (Option A) or position metrology (Option B). Absolute G measurements already reach about 10–30 ppm per apparatus [5], so a differential $\sigma(\delta) \sim$ 20–40 ppm is a realistic first goal.
- **Reaching 10^{-6}.** The full kill line for the dark-energy-tied framework needs about 20–30× beyond today's best. That would be a longer-term programme.

3 What each outcome would mean

- $|\delta| \approx 3 \times 10^{-4}$ **(source-dependent).** Laboratory G depends on the source acceleration. That excludes Newtonian gravity at laboratory scales, excludes the dark-energy-tied framework, and supports source-field-boost models.
- $|\delta| < 50\,\textbf{ppm}$. Source-field-boost models of the published size are excluded. The null predictions survive. The discrepancy between published laboratory G values then has to come from apparatus systematics.
- **Any intermediate value.** It would be the first measured acceleration dependence of laboratory G, and a strong result whatever the theory.

The outcome is informative either way, at modest cost: one additional source configuration on an apparatus that already exists.

4 Caveats

- The Quinn geometry used here (11 kg sources at about 0.21 m) and the Scarpa et al. numbers were transcribed from memory and from a summary of their paper. They are approximate; each group should recompute the predictions for its own geometry.
- The dark-energy-tied predictions come from a framework with a fitted coefficient ($\kappa = \frac{1}{2}$) and a bounded stiffness length. They are falsifiable predictions, not established physics.

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[1] R. Scarpa, R. Falomo & A. Treves, "Gravity anomaly from laboratory experiments to astrophysics", arXiv:2603.17770 (2026), using Klein's interpolation. [2] This programme, lane CFG772 (frozen criteria, script and outputs in the public repository), with the sealed kill line $|\Delta G/G| \geq 10^{-6}$. [3] Lane CFG770: the stiffened action against the 2026 Cassini quadrupole bound. [4] R. S. Park et al., arXiv:2602.17884 (2026). [5] T. Quinn et al., Phys. Rev. Lett. 111, 101102 (2013); CODATA 2018 G compilation. AI assistance: Anthropic's Claude.